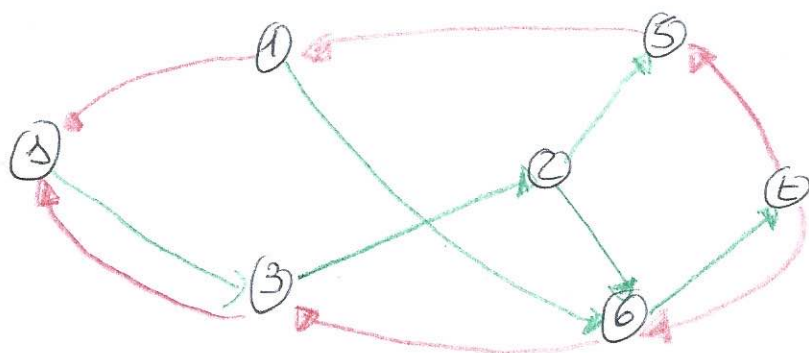


RESIDUAL GRAPH

①

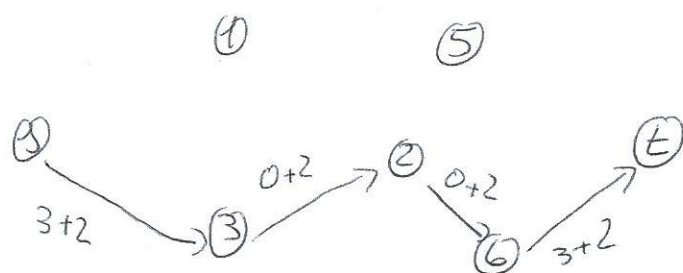


Path (with the minimum number of arcs) from s to t

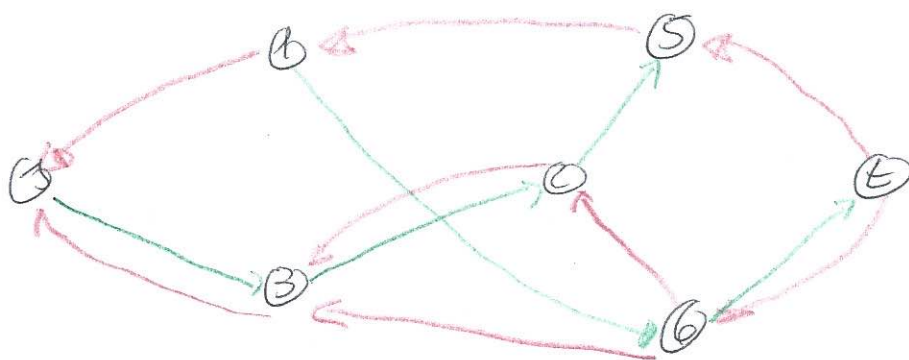
$s \xrightarrow{3} 3 \xrightarrow{2} 2 \xrightarrow{6} 6 \xrightarrow{t}$

$$\theta = \min\{10-3, 8-0, \boxed{2-0}, 10-3\} = 2$$

updated flow



new residual graph

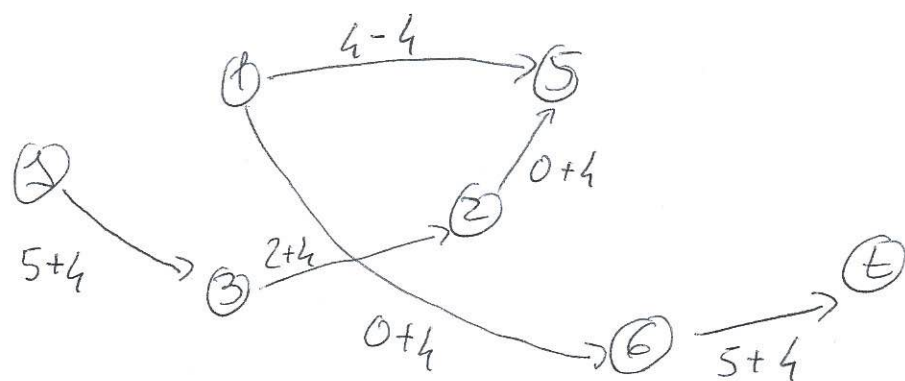


Path from s to t

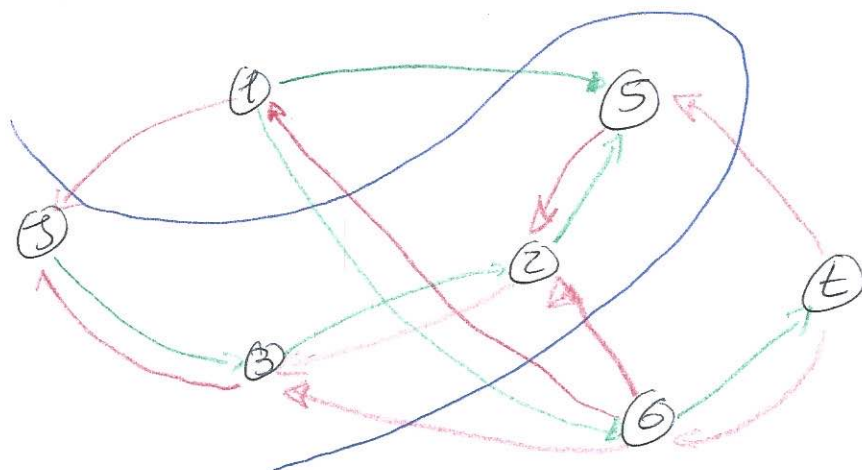
$s \xrightarrow{3} 3 \xrightarrow{2} 2 \xrightarrow{5} 5 \xrightarrow{1} 1 \xrightarrow{6} 6 \xrightarrow{t}$

$$\theta = \min\{10-5, 8-2, 5-0, \boxed{4}, 5-0, 10-5\}$$

updated flow



new residual graph



there is no path from s to t

cut $N_s = \{s, 3, 2, 5\}$ $N_t = \{1, 6, t\}$

Arc $(1, 5)$ goes from N_t to N_s (and $x_{15} = 0$)

If the capacity of $(1, 5)$ increases the maximum flow would not change.

The worst case complexity of $\bar{E}-K$ algorithm

is $O(m^2 u)$

where $m = |A|$ and $u = |U|$

EXERCISE 2

③

$$\begin{aligned} D) \quad \min \quad & -3y_1 + 4y_2 + 15y_3 \\ & -6y_1 + 2y_3 = -2 \\ & -2y_1 + y_2 + 3y_3 = 1 \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

CS equations

$$y_1(-6x_1 - 2x_2 + 3) = 0 \quad (1)$$

$$y_2(x_2 - 4) = 0 \quad (2)$$

$$y_3(2x_1 + 3x_2 - 15) = 0 \quad (3)$$

$$\text{solution } x' = \begin{bmatrix} -3/2 \\ 6 \end{bmatrix}$$

$$\text{from (2)} \quad 6 - 4 \neq 0 \Rightarrow y_2 = 0$$

$$\left. \begin{aligned} -6y_1 + 2y_3 &= -2 \\ -2y_1 + 3y_3 &= 1 \end{aligned} \right\} \quad y_1 = 4/7 \quad y_3 = 5/7$$

$$\text{complementary dual solution } y' = [4/7, 0, 5/7]$$

being ≥ 0 , it is feasible

x' violates constraint 2

$$6 \not\leq 4$$

x' is not primal feasible

x' and y' are not optimal

solution $x'' = \begin{bmatrix} 3/2 \\ 4 \end{bmatrix}$

(4)

from (1) $-6\left(\frac{3}{2}\right) - 2(4) + 3 = -9 - 8 + 3 \neq 0$

$\Rightarrow y_1 = 0$

$$\left. \begin{array}{l} 2y_3 = -2 \\ y_2 + 3y_3 = 1 \end{array} \right\} \quad \begin{array}{l} y_3 = -1 \\ y_2 = 4 \end{array}$$

complementary dual solution $y'' = [0, 4, -1]$

$y_3'' < 0$: y'' is infeasible

x'' satisfies all constraints

$\Rightarrow$ it is primal feasible

x'' and y'' are not optimal

EXERCISE 3

Since the upper bound does not increase ⑤
with the level of the enumeration tree, the
upper bound in node 5 is ≤ 87 (upper bound
of the closest ancestor)

The outcome of the upper bound computation
in node 7 that stops the overall computation
is a feasible solution ($ub = lb$) with value
 ≥ 87 that would make useless the explorations
of 5.

$$\text{Or } ub = lb \geq ub(5)$$

Ex 4

6

$$a^1, a^2, \dots, a^r \in \mathbb{R}^n$$

$$\text{cone}(a^1, a^2, \dots, a^r) =$$

$$\{x : x = \mu_1 a^1 + \mu_2 a^2 + \dots + \mu_r a^r, \mu_1, \dots, \mu_r \geq 0\}$$

Consider two points x and $y \in \text{cone}(a^1, a^2, \dots, a^r)$

$$x = \mu_1 a^1 + \dots + \mu_r a^r$$

$$y = \rho_1 a^1 + \dots + \rho_r a^r$$

and $\lambda \in [0, 1]$

the convex combination $\lambda x + (1-\lambda)y \in \text{cone}(a^1, \dots, a^r)$

Indeed

$$\lambda x + (1-\lambda)y = (\lambda\mu_1 + (1-\lambda)\rho_1)a^1 + \dots + (\lambda\mu_r + (1-\lambda)\rho_r)a^r$$

calling $\lambda_i = \lambda\mu_i + (1-\lambda)\rho_i$

$$\lambda x + (1-\lambda)y = \lambda_1 a^1 + \dots + \lambda_r a^r$$

being $\lambda_i \geq 0$

$$\lambda x + (1-\lambda)y \in \text{cone}(a^1, \dots, a^r)$$

in alternative.

Since a finitely generated cone is a polyhedral cone $Ax \leq 0$

then it is a convex set

Exercise 5

(7)

Sets

$\{1..K\}$ feeds
 $\{1..2\}$ raw materials
 $\{1..3\}$ bundles

Parameters

a_{ij} $i=1..K$ $j=1..2$ quantity of raw material j needed to produce 1 unit of feed i
 b_{hj} $h=1..3$ $j=1..2$ quantity of raw material j in bundle h
 c_h $h=1..3$ cost of bundle h
 g_i $i=1..K$ revenue per unit of feed i

Variables

$x_i \geq 0$ $i=1..K$ produced amount of feed i
 $y_h = \begin{cases} 1 & \text{if bundle } h \text{ is bought} \\ 0 & \text{otherwise} \end{cases}$ $h=1..3$

objective function

$$\max \sum_{i=1}^K g_i x_i - \sum_{h=1}^3 c_h y_h$$

resource availability constraints

$$\sum_{i=1}^K a_{ij} x_i \leq \sum_{h=1}^3 b_{hj} y_h \quad j=1..2$$

Variant

parameter: F_j $j=1..2$ cost for the inventory set up for raw material j

$$z_j = \begin{cases} 1 & \text{if raw material } j \text{ is present} \\ 0 & \text{otherwise} \end{cases}$$

coherence constraints

$$\sum_{h=1}^3 b_{hj} y_h \leq M z_j \quad (M \text{ big constant})$$

$j=1..2$

new objective function

$$\max \sum_{i=1}^K g_i x_i - \sum_{h=1}^3 c_h y_h - \sum_{j=1}^2 F_j z_j$$

EXERCISE 6

(8)

$$N = \{0, 1, \dots, u, u+1\}$$

$$A = \{ (i, j) : \sum_{h=i}^{j-1} P_{h+1} \leq U, i < j \}$$

the consumption does not exceed the tank capacity

$$l_{ij} = c_j \left(\sum_{h=i}^{j-1} P_{h+1} \right) \quad \begin{matrix} i=0 \dots u-1 \\ j=1 \dots u \end{matrix} \text{ and } (i, j) \in A$$

cost of refuelling

$$l_{iu+1} = 0$$

$$i=0 \dots u-1 \text{ and } (i, u+1) \in A$$

The graph is acyclic since if $(i, j) \in A$ then $i < j$

The most efficient algorithm is SPT_Acyclic $O(m)$
where $m = |A|$

AMPL exercise: solution

```
set I;  
set J;  
var x{I,J};  
param b{J};  
  
s.t. c1 {j in J: b[j] == 15}:  
      sum{i in I}x[i,j]  <= 5;
```